

$$\textcircled{1} \int \frac{e^x}{e^x + 1} dx$$

$$u = ax + b$$

$$du = a dx$$

$$\frac{du}{a} = dx$$

$$4) \int \frac{1}{ax+b} dx = \int (ax+b)^{-1} dx$$

$$= \ln |ax+b| + C$$

$$\int \frac{1}{u} \frac{du}{a} = \int (au)^{-1} du = \ln |au| + C$$

$$\neq \ln |a(ax+b)| + C$$

$$= \frac{1}{a} \int (u)^{-1} du = \frac{1}{a} \ln |ax+b| + C$$